



Preparation of fibrous polymer grafted magnetic chitosan beads and modified with hydrazone and imidazole groups: isolation of U(VI) ions from seawater

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Abstract

The magnetic chitosan beads were prepared and grafted with glycidylmethacrylate, then diaminomelanonitrile (DAMN) was attached. The nitrile groups of DAMN were reacted with hydrazine hydrate hydrochloride and hydrazine hydrate-ethylene diamine to obtain hydrazine (HYD) and imidazole (IMDZ) groups, respectively. Both adsorbents removed uranyl ions from the aqueous medium and seawater. The adsorption capacities of the $\text{Fe}_3\text{O}_4@\text{CHT}@p(\text{GMA})\text{-HYD}$ and $\text{Fe}_3\text{O}_4@\text{CHT}@p(\text{GMA})\text{-IMDZ}$ beads were 328.4 and 434.7 mg/g, respectively. Both adsorbents have high performance for adsorbing U(VI) ions from seawater. The synthesized adsorbents were used to isolate U(VI) ions from seawater in the presence of various ions.

Keywords Chitosan-based adsorbent · Surface modification · Adsorption · Uranium · Isotherms and Kinetics

Introduction

Developing clean and renewable energy has become one of the global concerns [1–4], and long-term sustainable global energy supply has become one of the most important problems for humanity [5, 6]. The present energy resources, such as petrol, oil, natural gas, and mineral gases, are restricted and can be run out. Therefore, with the rapid increase in global energy demands, nuclear power has appeared as a viable solution to mitigate greenhouse gas emissions and increase energy intensity [7–9]. In the nuclear industry,

U(VI) is a crucial element widely distributed in the Earth's crust and especially abundant in seawater. The known reserves of uranium ore on land can be inadequate to meet the future demands of nuclear power plants. The estimated amounts of uranium in the seas are approximately 4.5 billion tons, around a thousand times more than the total land-based reserves [10, 11]. Therefore, selective and easy isolation of U(VI) ions from seawater is essential for the sustainable supply of uranium to nuclear energy resources [10–12]. Many methods have been reported for the isolation of U(VI) ions from seawater, such as chemical precipitation, photo-catalysis, membrane filtration, and adsorption [13–17]. Among these techniques, adsorption can be considered a sustainable method for U(VI) removal from seawater due to its low cost, large-scale applicability, high efficacy, and easy operation [18–22]. Numerous types of sorbents have been used for the adsorption of inorganic and organic from environment, including active carbon [23], composite materials [24, 25], cellulose [17, 26], chitosan [27, 28], metal–organic framework [29], layered double zinc and aluminum hydroxide [30–32], hydrogels [33], various polymeric materials [34–36], biosorbents [24, 37], and magnetic adsorbents [38–41]. However, these prepared adsorbents have some advantages and disadvantages, and thus, to prepare reliable adsorbents with desired properties that are cost-effective, easy to produce, highly durable, and possess a high

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adsorption capacity under various experimental conditions. It should be noted that adsorbents containing amine groups are important for rapid adsorption because they easily interact with metal ions. Thus, they could effectively remove uranyl ions from seawater [7, 33, 38, 42–44]. In previous studies, researchers preferred to develop structures containing amine groups, especially polyamine and imidazole groups, when designing new adsorbents. For example, Bayramoglu et al. prepared agmatine ligand-modified magnetic alginate beads, and the adsorption capacity of the adsorbent for U(VI) was 451.4 mg/g. Bayramoglu and Arica synthesized poly amidoxime group functionalized magnetic adsorbent and removed U(VI) ions from aqueous solutions. The maximum adsorption pH was observed at 6.0. Yang et al. [27] prepared amidoxime-modified chitosan adsorbent, uranium removal was studied from aqueous solution and seawater, and the maximum uranium adsorption capacities were 596 mg/g, and 243.2 µg/g, respectively. Ivanov et al. (2024) prepared Fe₃O₄ coated by Zn and Al layered double hydroxide for U(VI) removal, and the adsorption exchange capacity towards U(VI) ions was reported as 268.65 mg/g [32]. Additionally, adsorbents should be easily separable from the adsorption medium and not generate secondary pollutants. Magnetic materials can be simply removed from the solutions by using an external magnet. For this reason, magnetic adsorbents have received significant attention for removing metal ions from solutions [13, 32, 38]. A short time since nanotechnology has been applied in all branches of the sciences. With this understanding, new-generation adsorbents were prepared by grafting comb-type polymer chains in adsorption technology. In this concept, magnetic composite material can be grafted with a comb-type polymer to improve its adsorption performance using different functional polymers. Among the grafting methods, redox polymerization grafting is an easy polymerization method [35], which does not involve severe experimental conditions. In the grafting polymerization method, GMA is a widely used monomer, and its polymer has reactive epoxy groups. It can be utilized for coupling reactions with amine and thiol groups of the ligand molecules. Furthermore, the graft polymerization method generates fibrous polymer on the magnetic adsorbent, and various ligands can be attached to the fibrous polymer. This kind of adsorbent offers a high surface area [1]. In this study, magnetic chitosan beads were prepared via the phase inversion method and functionalized with hydrazone and imidazole groups. The synthesized adsorbents were characterized by FT-IR, SEM, BET, and analytical methods. The adsorption capacities of both adsorbents (i.e., Fe₃O₄@CHT@p(GMA)-HYD and Fe₃O₄@CHT@p(GMA)-IMDZ beads) for uranium under various experimental conditions were investigated in batch and continuous systems. The kinetic and thermodynamic models evaluated the adsorption mechanism between U(VI)

ions and the adsorbents. Finally, the performance of the as-synthesized adsorbent was tested for adsorption of U(VI) ions from seawater obtained from the Black Sea and the Mediterranean Sea.

Materials and methods

Materials

Glycidyl methacrylate (GA), diaminomaleonitrile (DAMN, 98%), ethylenediamine, hydrazine hydroxylamine (NH₂OH), and chitosan (high molecular weight, deacetylation ≥ 75%), were obtained from Sigma-Aldrich (Hamburg, Germany). U(VI) solution was prepared by dissolving uranyl acetate (UO₂(CH₃COO)₂·2H₂O) in deionized water and supplied from Honeywell Riedel-de-Haen AG (Germany). FeCl₃, FeSO₄·7H₂O, isopropanol (IPA), and all other chemicals were of analytical grade and purchased from Merck (Darmstadt, Germany).

Preparation of magnetic nanoparticles

The magnetic nanoparticles were synthesized via thermal co-precipitation reaction of Fe(III) and Fe(II) ions in a 2:1 molar ratio. For this purpose, FeCl₃ (6.5 g, 0.04 mol) and FeSO₄·7H₂O (5.6 g, 0.02 mol) were separately dissolved in 200 mL of distilled water. Then, the FeCl₃ (50 mL) and FeSO₄ (50 mL) solutions were transferred to a reaction vessel, and the temperature was raised to 45 °C. After reaching the thermal equilibrium, ammonia solution (50 mL, 25% w/v) was added dropwise. The reaction temperature was set to 75 °C and allowed to proceed for two hours under a nitrogen atmosphere. After this period, the reaction temperature was raised to 85 °C and continued for one hour. After the reaction period, the magnetic nanoparticles (Fe₃O₄) were separated using an external magnet and washed several times with distilled water to remove impurities. Then, they were dried in a vacuum oven under reduced pressures at 45 °C.

Synthesis of magnetic chitosan beads

Magnetic chitosan beads were synthesized via the phase inversion method, as described previously [28]. Briefly, chitosan solution was prepared by adding 0.4 g of chitosan flake in 10% acetic acid solution (v/v, 40 mL). Then, 1.0 g of the above magnetic nanoparticles were added and magnetically mixed until a homogeneous solution was obtained. This mixture was dropped into NaOH solution (2.5 mol/L) using a syringe pump through an injection needle while continuously stirring the solution. The resulting Fe₃O₄@CHT spheres were washed with distilled water and stored.

Grafting of magnetic chitosan beads with GMA and modification into hydrazone and imidazole groups

The magnetic chitosan beads were grafted with glycidylmethacrylate. The grafting reaction time was realized at different reaction times between 3.0 and 10 h to obtain the desired amount of GMA grafting density. Briefly, the $\text{Fe}_3\text{O}_4\text{@CHT}$ beads (~ 300 mg) were transferred to a reaction vessel, then isopropanol (20 mL) and deionized water (20 mL) were added and heated to 50°C under a nitrogen atmosphere. For the grafting reaction, GMA (2.0 mL) and KMnO_4 (20 mL, 2.0 mmol/L) were added to this mixture, and the reaction was continued for 10 h. After this period, $\text{Fe}_3\text{O}_4\text{@CHT@p(GMA)}$ beads were washed with distilled water and ethanol 70% (v/v) to remove reaction impurities and then stored in ethanol (70%, v/v).

Diaminomaleonitrile (DAMN) ligand was covalently attached to the $\text{Fe}_3\text{O}_4\text{@CHT@p(GMA)}$ beads via the reaction between the epoxy groups of p(GMA) and the amino groups of DAMN. Briefly, DAMN ligand (0.3 g) was dissolved in ethanol (100 mL, pH 9.0), and $\text{Fe}_3\text{O}_4\text{@CHT@p(GMA)}$ beads 0.5 g were added. The reaction mixture was rotated at 150 rpm and 60°C for 12 h. After the reaction, $\text{Fe}_3\text{O}_4\text{@CHT@p(GMA)-DAMN}$ beads were washed with ethanol (70%, v/v) and distilled water and stored at room temperature.

For hydrazone group modification, $\text{Fe}_3\text{O}_4\text{@CHT@p(GMA)-DAMN}$ beads (0.5) were transferred into a reaction medium containing hydrazine hydrate (15 mL) and ethanol (5 mL) at pH 8.0. The reaction medium was refluxed at 85°C for 6.0 h. At the end of the reaction period, the hydrazine-modified beads (i.e., $\text{Fe}_3\text{O}_4\text{@CHT@p(GMA)-HYD}$) were cleaned with ethanol and distilled water and then stored at room temperature. For imidazole group generation on the $\text{Fe}_3\text{O}_4\text{@CHT@p(GMA)-HYD}$ beads were reacted with ethylene diamine (EDA). An EDA solution (33%, 80 mL) containing 0.1 g of sulfur was prepared for this. Then, the solution temperature was raised to 90°C , and the dried $\text{Fe}_3\text{O}_4\text{@CHT@p(GMA)-HYD}$ beads were added to the reaction medium. The reaction was continued at $95^\circ\text{C} \pm 5^\circ\text{C}$ for 7.0 h. After the reaction period, the imidazole-modified beads ($\text{Fe}_3\text{O}_4\text{@CHT@p(GMA)-IMDZ}$) were washed with deionized water at 50°C and then stored in ethanol: water mixture (70:30, v/v) at room temperature.

Characterization studies

The ATR-FTIR spectra of magnetic Fe_3O_4 nanoparticles, $\text{Fe}_3\text{O}_4\text{@CHT}$, $\text{Fe}_3\text{O}_4\text{@CHT@p(GMA)}$, $\text{Fe}_3\text{O}_4\text{@CHT@p(GMA)-HYD}$, and $\text{Fe}_3\text{O}_4\text{@CHT@p(GMA)-IMDZ}$, and beads were obtained using an FTIR spectrometer (Nicolet iS5, Thermo Fisher Scientific, U.S.A.). The

surface morphology of the as-prepared magnetic beads was analyzed using scanning electron microscopy (SEM, FEI QUANTA 250 FEG, Oregon, USA). The magnetic properties of the Fe_3O_4 nanoparticles, $\text{Fe}_3\text{O}_4\text{@CHT}$, and $\text{Fe}_3\text{O}_4\text{@CHT@p(GMA)}$ beads were studied using a vibrating sample magnetometer (VSM, 880; ADE Technologies, USA). The surface areas and average pore sizes of the Fe_3O_4 nanoparticles, $\text{Fe}_3\text{O}_4\text{@CHT}$, and $\text{Fe}_3\text{O}_4\text{@CHT@p(GMA)}$ beads were obtained by the Brunauer–Emmett–Teller (BET) analysis. The N_2 adsorption/desorption experiments were studied using ASAP2020 HD88. The amount of epoxy groups on the $\text{Fe}_3\text{O}_4\text{@CHT@p(GMA)}$ beads was determined using the pyridine-HCl method described previously [1]. An ICP-MS spectrometer Thermo Fisher (iCAP TQe00061, Scientific, USA) with a plasma generator and the power output 1550 was used to determine metals. The as-prepared materials' x-ray diffraction (XRD) patterns were obtained using Cu-K α radiation, between 2θ of 20 and 80° , with a 0.1 increment and 2° min^{-1} scan speed (MiniFlex 600, Rigaku).

Adsorption studies

The adsorption performances of the $\text{Fe}_3\text{O}_4\text{@CHT@p(GMA)-HYD}$ and $\text{Fe}_3\text{O}_4\text{@CHT@p(GMA)-IMDZ}$ were studied batch-wise. The adsorption experiments were conducted in a vial containing 20 mg of adsorbents and 30 mL of uranyl ions solution. The vials were placed on a shaker and rotated at 200 rpm and $25 \pm 2^\circ\text{C}$.

The influence of pH on the adsorption performance was examined between pH 3.0 and 8.0, and U(VI) ions concentration was 200 mg/L. The effect of adsorbent amount on the adsorption performance was studied with different amounts of adsorbent in the range of 0.2–1.0 g/L with a uranium ions concentration of 200 mg/L. The impact of the initial concentration of uranium ions on adsorption capacity was investigated by varying the concentration of U(VI) ions between 1.0 and 400.0 mg/L. The effect of contact time on adsorption was investigated at a concentration of 400 mg/L of uranium ions for 180 min, and the sample was taken at predetermined time intervals. The adsorption performance of the adsorbents was also studied in synthetic seawater. One liter of synthetic seawater comprised of the following ingredients: NaCl: 24.53 g, MgCl_2 : 5.20 g, CaCl_2 : 1.16 g, KCl: 0.695 g, CuCl_2 : 20 mg, NiCl_2 : 20 mg and PbCl_2 20 mg) and various amount of uranium ions was added to this solution between 0.5 and 10.0 mg/L. The amount of adsorbed U(VI) ions per unit of adsorbent [mg U(VI)/g adsorbent] was calculated by using the following equation:

$$q_e = (C_0 - C_e/m) V \quad (1)$$

where q_e is the amount of U(VI) ion adsorbed onto the unit mass of adsorbent (mg/g), C_0 and C_e are the concentrations

of the U(VI) ion before and after adsorption (mg/L), V is the volume of the solution (L), and m is the amount of the adsorbent (g).

Competing adsorption studies from binary mixture

The selectivity of the $\text{Fe}_3\text{O}_4@\text{CHT}@p(\text{GMA})\text{-HYD}$ and $\text{Fe}_3\text{O}_4@\text{CHT}@p(\text{GMA})\text{-IMDZ}$ beads towards U(VI) ions (ionic radius: 138 pm) in the presence of competitive ions Cu(II) (ionic radius: 71 pm) and/or Ni(II) (ionic radius: 73 pm) was evaluated in a batch system. The metal ions solutions were prepared from $\text{Cu}(\text{NO}_3)_2 \cdot 3\text{H}_2\text{O}$, $\text{Ni}(\text{NO}_3)_2 \cdot 6\text{H}_2\text{O}$ and $\text{Pb}(\text{NO}_3)_2$. A solution (20 mL) containing 1.0 mmol/L U(VI) ions and Cu(II) ions, Ni(II) or Pb(II) ions each of them 20 mmol/L was contacted with 1.0 g/L of $\text{Fe}_3\text{O}_4@\text{CHT}@p(\text{GMA})\text{-HYD}$ and $\text{Fe}_3\text{O}_4@\text{CHT}@p(\text{GMA})\text{-IMDZ}$ beads at pH 6.0 and 25 °C in the vial stirred at 200 rpm. After the adsorption period, the remaining concentration of each metal ion in the test solution was determined, as described above.

Removal of U(VI) ions from seawater in the continuous system

Removal of U(VI) ions from seawater in a continuous system was studied with $\text{Fe}_3\text{O}_4@\text{CHT}@p(\text{GMA})\text{-IMDZ}$ beads. The adsorbent was packed in a packed bed reactor (length: 14 cm; radius: 0.8 cm; volume: 28 mL), and the reactor was fed separately with the Black Sea water and the Mediterranean Sea water samples from the bottom inlet part at a flow rate of 100 mL/h with a peristaltic pump (model

IPC, Ismatec, Düsseldorf, Germany). The reactor was packed approximately with 3.8 g adsorbent and operated at room temperature (approximately 25 °C) for 24 h.

Desorption studies

Adsorbed U(VI) ions were desorbed from the adsorbent by using 0.1 mol/L nitric acid solution. The $\text{Fe}_3\text{O}_4@\text{CHT}@p(\text{GMA})\text{-HYD}$ and $\text{Fe}_3\text{O}_4@\text{CHT}@p(\text{GMA})\text{-IMDZ}$ beads were located in the desorption medium and stirred at 200 rpm at room temperature for 2.0 h. To determine the reusability of $\text{Fe}_3\text{O}_4@\text{CHT}@p(\text{GMA})\text{-HYD}$ and $\text{Fe}_3\text{O}_4@\text{CHT}@p(\text{GMA})\text{-IMDZ}$ beads successive adsorption/desorption rounds were repeated five times by using the same adsorbents. The desorption ratio was calculated from the amount of U(VI) ions released by the adsorbent and the final concentration of U(VI) ions in the adsorption medium.

Results and discussion

Characterization studies

A two-step process involved preparing $\text{Fe}_3\text{O}_4@\text{CHT}@p(\text{GMA})\text{-HYD}$, and $\text{Fe}_3\text{O}_4@\text{CHT}@p(\text{GMA})\text{-IMDZ}$ beads. $\text{Fe}_3\text{O}_4@\text{CHT}$ beads were prepared from magnetic nanoparticles and chitosan via the phase inversion method in the first step. The second step consisted of the graft polymerization of GMA monomer via redox polymerization in the presence of KMnO_4 . The schematic representation of the preparation of adsorbents is presented in Fig. 1. The grafting

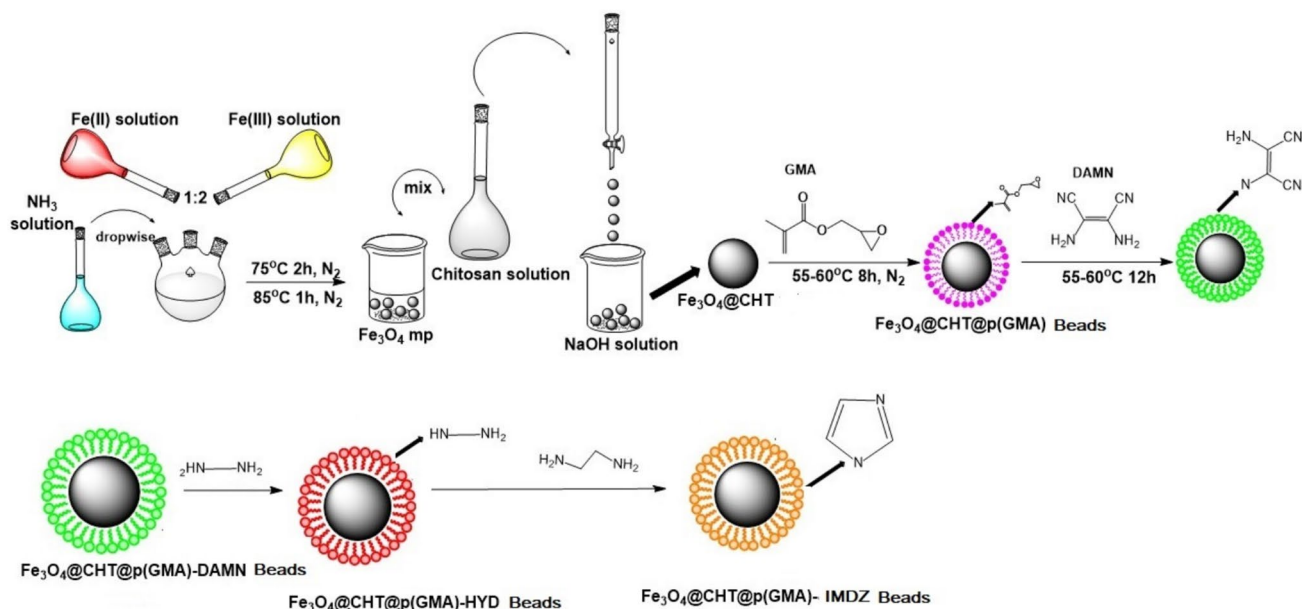


Fig. 1 Schematic representation of preparations of hydrazide and imidazole-modified adsorbents

reaction time was optimized to obtain high adsorption performance towards uranyl ions, and the polymerization time was realized up to 10 h. The amount of grafted p(GMA) on the magnetic chitosan beads was detected by the pyridine-HCl method. The functional epoxy group content of the beads was found to be between 0.63 and 1.87 mmol/g beads.

The ATR-FTIR spectra of Fe_3O_4 nanoparticles, CHT, Fe_3O_4 @CHT, Fe_3O_4 @CHT@p(GMA), Fe_3O_4 @CHT@p(GMA)-HYD, and Fe_3O_4 @CHT@p(GMA)-IMZD beads are presented in Fig. 2. The characteristic peak at 599 cm^{-1} corresponded to the Fe–O stretching vibration of iron oxide nanoparticles. The presence of this peak confirmed the successful synthesis of Fe_3O_4 nanoparticles (Fig. 2A). The IR spectrum of chitosan exhibited characteristic peaks at 3354 cm^{-1} due to –NH and –OH stretching vibrations. The peak at 1578 cm^{-1} could have resulted from the $-\text{NH}_2$ stretching vibration of chitosan (Fig. 2B). The FTIR spectra were also obtained to confirm the presence of Fe_3O_4 nanoparticles in the chitosan beads at 599 cm^{-1} . The

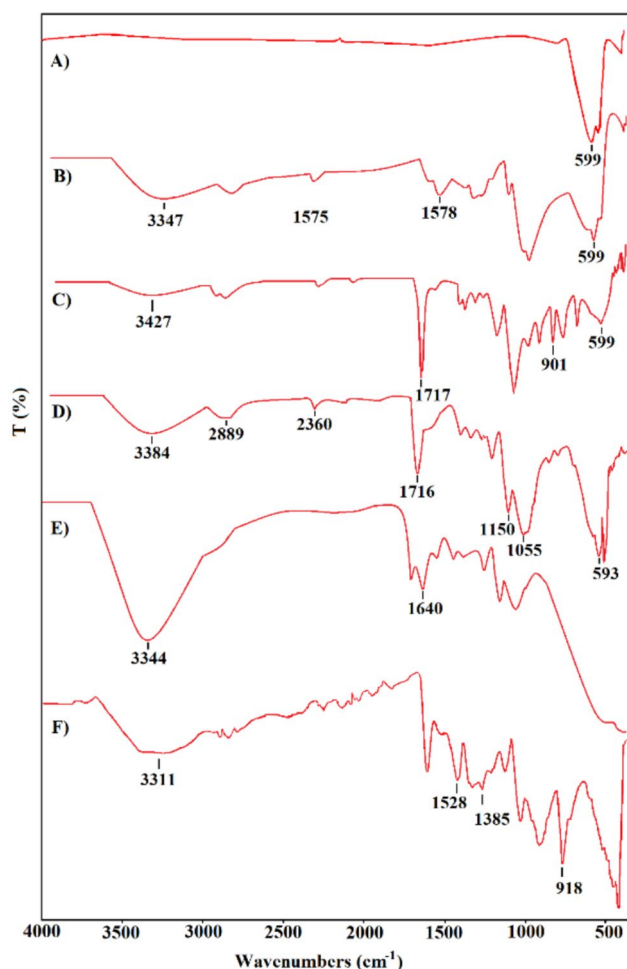


Fig. 2 FT-IR Spectra of **A** Fe_3O_4 , **B** Fe_3O_4 @CHT, **C** Fe_3O_4 @CHT-g-p(GMA), **D** Fe_3O_4 @CHT-g-p(GMA)-DAMN, **E** Fe_3O_4 @CHT-g-p(GMA)-HYD, and **F** Fe_3O_4 @CHT-g-p(GMA)-IMDZ

IR spectrum of Fe_3O_4 @CHT displayed a band at 3347 cm^{-1} due to –NH or –OH stretching vibrations. Similar peaks (namely amide I, amid II, and Fe–O) were observed in this spectrum as in the case of pure chitosan. The FT-IR spectra of the Fe_3O_4 @CHT@p(GMA) beads with characteristic peaks of epoxy and carbonyl groups at 901 cm^{-1} and at 1717 cm^{-1} were observed, respectively (Fig. 2C). The FTIR spectrum for Fe_3O_4 @CHT@p(GMA)-DAMN is presented in Fig. 2D. As observed from this spectrum, the band is due to antisymmetric stretching vibrations of the NH_2 groups observed at 3384 cm^{-1} . The characteristic stretching vibration peak of $\text{C}\equiv\text{N}$ was observed at 2360 cm^{-1} . The band at 1716 cm^{-1} could be assigned for the NH_2 scissoring vibrations. The FT-IR spectra of Fe_3O_4 @CHT@p(GMA)-HYD beads showed peaks at around 3344 cm^{-1} and 1640 cm^{-1} , corresponding to $-\text{NH}_2$ stretching vibration and N–H deformation, respectively, indicating the incorporation of amino groups from hydrazine into the structure of nitrile groups of DAMN on the beads (Fig. 3E). The FT-IR spectra of the Fe_3O_4 @CHT@p(GMA)-IMDZ exhibited new bands at 3311 cm^{-1} for N–H stretching vibration. The peaks at 1528 and 1385 cm^{-1} could be assigned for $-\text{NH}_2$ and $-\text{CN}$ stretching vibrations, confirming the formation of imidazole groups onto the HYD-modified beads after reaction with ethylene diamine (Fig. 2F). Furthermore, the new peak observed at 918 cm^{-1} is related to the wagging of the primary and secondary amines of the imidazole ring. Thus, as observed changes in the spectra, we conclude that the attached DAMN ligands were successfully converted to hydrazone and imidazole groups on the magnetic beads.

The SEM image can provide information on the size, shape, and surface morphology of the as-prepared adsorbent material. The surface morphologies of the Fe_3O_4 @CHT and Fe_3O_4 @CHT@p(GMA) beads were studied using scanning electron microscopy (Fig. 3). As observed in Fig. 3A, a sphere-like structure was obtained for Fe_3O_4 @CHT beads, and at high magnification, the surface of the Fe_3O_4 @CHT beads was found to be a highly irregular structure (Fig. 3B). After grafting with p(GMA), a structural change in the surface morphology of Fe_3O_4 @CHT@p(GMA) beads was observed and suggested that the grafting of the surface of the Fe_3O_4 @CHT beads with p(GMA) was successfully achieved (Fig. 3C). The surface of the Fe_3O_4 @CHT@p(GMA) beads was highly porous. Thus, the porous surface can provide a large surface area, and as a result, the overall surface area can be significantly enhanced. Grafting with p(GMA) may increase the contact surface area and adsorption capacity of uranium ions in the solution.

The magnetic properties of the as-prepared materials were obtained using a VSM. As shown in Fig. 4, the magnetic hysteresis curves of the Fe_3O_4 nanoparticles, Fe_3O_4 @CHT, and Fe_3O_4 @CHT-g-p(GMA) beads were attained S-shaped over the elaborated magnetic field, representing

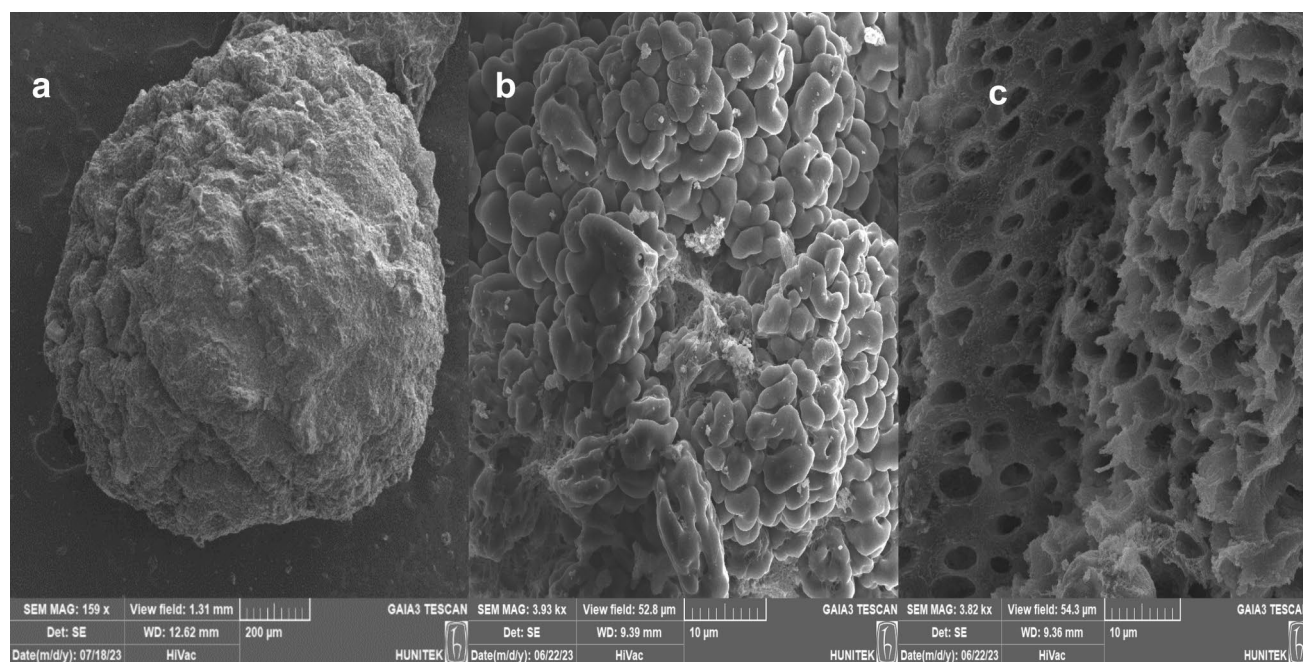


Fig. 3 SEM images of **a** Fe₃O₄@CHT beads at ~160X, **b** Fe₃O₄@CHT beads at ~4kX, **c** Fe₃O₄@CHT@p(GMA) beads at ~4kX

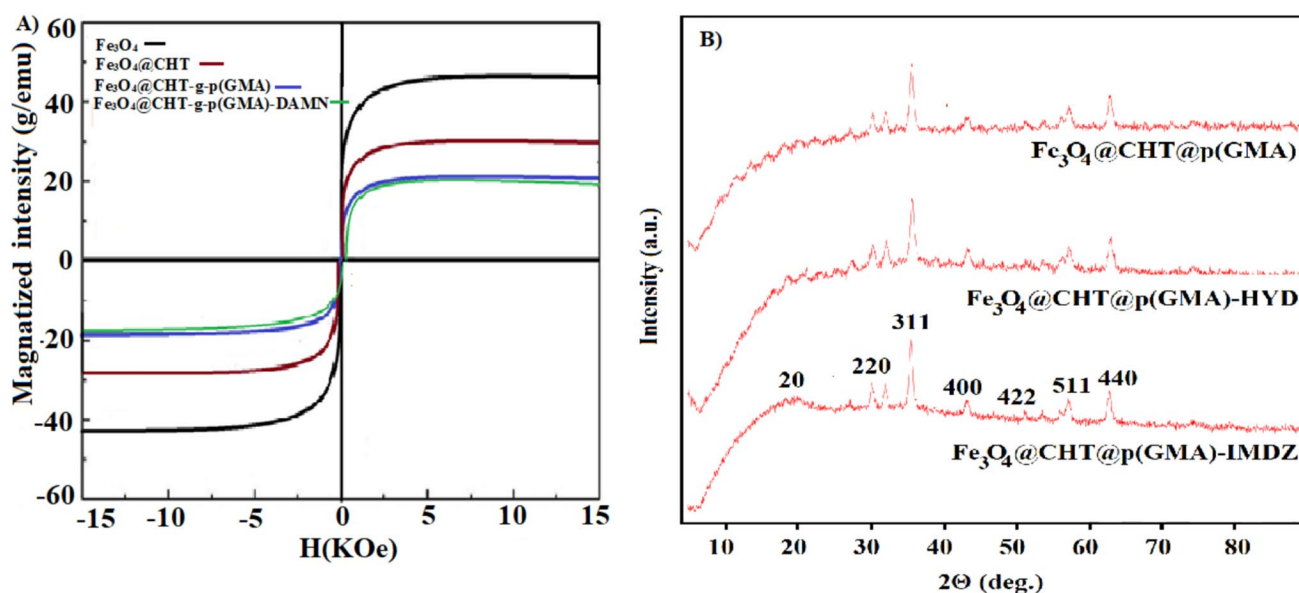


Fig. 4 VSM profiles of Fe₃O₄, Fe₃O₄@CHT, Fe₃O₄@CHT-g-p(GMA) and Fe₃O₄@CHT-g-p(GMA)-DAMN (A); XRD patterns of Fe₃O₄@CHT-g-p(GMA), Fe₃O₄@CHT-g-p(GMA)-HYD, and Fe₃O₄@CHT-g-p(GMA)-IMDZ (B)

that the tested materials had magnetic properties (Fig. 4). Fe₃O₄ nanoparticles, Fe₃O₄@CHT, and Fe₃O₄@CHT-g-p(GMA) beads revealed durable magnetism with low remanence and coercivity, indicating these magnetic materials were easily removed from the adsorption medium by just placing in a magnetic field after laden with pollutants. The magnetic performances of the Fe₃O₄ nanoparticles,

Fe₃O₄@CHT, Fe₃O₄@CHT-g-p(GMA), and Fe₃O₄@CHT-g-p(GMA)-DAMN beads were found to be 41.3, 32.7, 18.9 and 18.4 emu/g, respectively. The magnetic properties of the Fe₃O₄@CHT and Fe₃O₄@CHT-g-p(GMA) decreased compared to Fe₃O₄ nanoparticles as the amount of non-magnetic materials increased. The modification of the DAMN ligand into hydrazone and imidazole groups was

not caused any change in the magnetic properties of the adsorbents.

The XRD patterns of $\text{Fe}_3\text{O}_4@\text{CHT-g-p(GMA)}$, $\text{Fe}_3\text{O}_4@\text{CHT-g-p(GMA)-HYD}$, and $\text{Fe}_3\text{O}_4@\text{CHT-g-p(GMA)-IMDZ}$ are presented in Fig. 7B. From this figure, diffraction peaks for the polymers grafted magnetic particles were detected at around $2\theta = 20^\circ$. The observed peaks in the XRD pattern of the as-prepared materials were pertinent to the crystalline phase of the beads. By matching with the reference peaks of the magnetic materials, six diffraction peaks around $2\theta = 32.7^\circ$, 36.8° , 43.5° , 47.2° , 58.3° , and 64.8° were associated with the face-centered cubic structure of magnetic particles within the bead structure. The reflection peaks agreed well with the X-ray diffraction data cards (ICDD number 88–0315) with six diffraction peaks: 220, 311, 400, 422, 511, and 440, respectively. These reflection peaks agree well with the X-ray diffraction data cards (ICDD number 88–0315) with six diffraction peaks: 220, 311, 400, 422, 511, and 440, respectively. As expected, the magnetic properties of the $\text{Fe}_3\text{O}_4@\text{CHT-g-p(GMA)}$ was not affected by the modification reactions as observed from XRD data.

The average bead size distribution of the $\text{Fe}_3\text{O}_4@\text{CHT}$ and $\text{Fe}_3\text{O}_4@\text{CHT-g-p(GMA)}$ was found to be 1.1 ± 0.2 and 1.3 ± 0.3 mm, respectively. The DAMN ligand attachment and modification reactions were not significantly affected the beads size distribution. The measurements of surface area and pore size distributions of the $\text{Fe}_3\text{O}_4@\text{CHT}$, $\text{Fe}_3\text{O}_4@\text{CHT-g-p(GMA)}$, $\text{Fe}_3\text{O}_4@\text{CHT-g-p(GMA)-DAMN}$, $\text{Fe}_3\text{O}_4@\text{CHT-g-p(GMA)-HYD}$ and $\text{Fe}_3\text{O}_4@\text{CHT-g-p(GMA)-IMDZ}$ beads samples were studied low-temperature nitrogen adsorption–desorption isotherms. The surface areas of the $\text{Fe}_3\text{O}_4@\text{CHT}$ and $\text{Fe}_3\text{O}_4@\text{CHT-g-p(GMA)}$ beads were obtained with nitrogen adsorption/desorption experiments. The surface area of the $\text{Fe}_3\text{O}_4@\text{CHT}$ was 1.93 g/m^2 . On the other hand, the surface area of the $\text{Fe}_3\text{O}_4@\text{CHT-g-p(GMA)}$ beads was 5.96 g/m^2 , and the area was increased by about three folds after grafting with p(GMA), the average pore size of the $\text{Fe}_3\text{O}_4@\text{CHT-g-p(GMA)}$ was $1.85 \text{ }\mu\text{m}$. Thus, a high surface area and porous surface were generated on the $\text{Fe}_3\text{O}_4@\text{CHT}$ beads via grafting with p(GMA). The results showed that the surface area of the $\text{Fe}_3\text{O}_4@\text{CHT-g-p(GMA)}$ beads considerably increased after grafting with p(GMA) polymer. The surface area of the $\text{Fe}_3\text{O}_4@\text{CHT-g-p(GMA)-DAMN}$, $\text{Fe}_3\text{O}_4@\text{CHT-g-p(GMA)-HYD}$, and $\text{Fe}_3\text{O}_4@\text{CHT-g-p(GMA)-IMDZ}$ beads were found to be 4.71, 4.18 and 3.94 mg/g beads, and average pore sizes were 1.78, 1.66 and 1.48 mm , respectively. The surface areas and average pore sizes of the DAMN ligand attached and modified beads decreased compared to bare $\text{Fe}_3\text{O}_4@\text{CHT-g-p(GMA)}$ beads; these decreases might be due to the accumulation of ligand molecules on and around the surface of the pores, eventually leading to the blockage of the pore mouth.

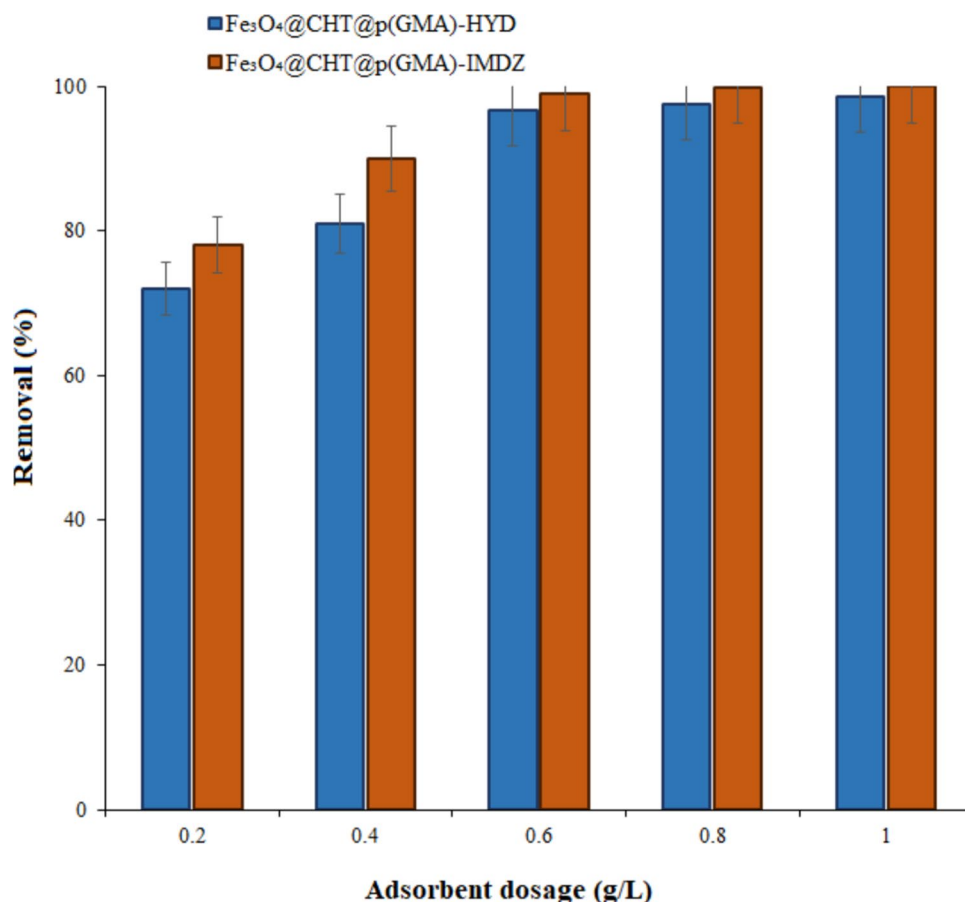
Effect of adsorbent dose on adsorption efficiency of U(VI) ions

The amount of adsorbent in the medium is a significant factor in pollutant removal studies. The adsorption of U(VI) onto $\text{Fe}_3\text{O}_4@\text{CHT@p(GMA)-HYD}$ and $\text{Fe}_3\text{O}_4@\text{CHT@p(GMA)-IMDZ}$ beads was studied at different amounts of adsorbent dose and changed between 0.2 and 1.0 g/L with an initial concentration of 200 mg/L . The adsorption efficiency of adsorbents for U(VI) under different adsorbent amounts is presented in Fig. 5. As seen in this figure, the amount of adsorbed U(VI) progressively increased with the increase of the adsorbent dosage. When the amount of adsorbent was 0.6 g/L , the removal percentage of $\text{Fe}_3\text{O}_4@\text{CHT@p(GMA)-HYD}$ and $\text{Fe}_3\text{O}_4@\text{CHT@p(GMA)-IMDZ}$ beads reached 96.7% and 98.9%, respectively. Then, the removal percentage did not enhance meaningfully with increasing the adsorbent dosage. This could be because the available binding sites on the increasing amount of adsorbent could not be saturated with constant uranyl ions concentration in the solution. Thus, the amount of adsorbed U(VI) ions by each tested adsorbent was sensible with the residual amount of uranyl ions in the solution, causing a quiet adsorption percentage. As the adsorbent dosage was enhanced, the amount of adsorbed uranyl ions on the unit of the adsorbent dose gradually reduced due to a decrease in the amount of U(VI) ions in the unit adsorbent mass. For the best results in the remaining studies, 0.6 g/L of $\text{Fe}_3\text{O}_4@\text{CHT@p(GMA)-HYD}$ or $\text{Fe}_3\text{O}_4@\text{CHT@p(GMA)-IMDZ}$ beads were used in the U(VI) ions adsorption test.

Effect of pH on adsorption performance of $\text{Fe}_3\text{O}_4@\text{CHT@p(GMA)-HYD}$ or $\text{Fe}_3\text{O}_4@\text{CHT@p(GMA)-IMDZ}$ beads

The pH value of the aqueous solution is crucial in regulating the uranium species and the surface properties of the adsorbent, thereby influencing the uranium adsorption capacity [9]. Studies on the effect of pH on the adsorption of U(VI) ions onto $\text{Fe}_3\text{O}_4@\text{CHT@p(GMA)-HYD}$ and $\text{Fe}_3\text{O}_4@\text{CHT@p(GMA)-IMDZ}$ beads were carried out in the pH range of 3.0–8.0. Moreover, control experiments without adsorbent were realized, and no precipitation of uranyl ions on the vial walls and any surface was observed at the studied pH range. The maximum uranyl ions adsorption on the $\text{Fe}_3\text{O}_4@\text{CHT@p(GMA)-HYD}$ and $\text{Fe}_3\text{O}_4@\text{CHT@p(GMA)-IMDZ}$ beads was observed between pH 6.0 and 7.0 (Fig. 6A). As observed from this figure, as the solution pH increased, the adsorption capacities of the tested adsorbents increased to pH 7.0 and then decreased. The protonation of hydrazone and imidazole groups of both adsorbents resulted in a positive surface charge on both adsorbents under acidic conditions. The pKa values of

Fig. 5 Adsorption performance of $\text{Fe}_3\text{O}_4@\text{CHT}@p(\text{GMA})\text{-HYD}$ and $\text{Fe}_3\text{O}_4@\text{CHT}@p(\text{GMA})\text{-IMDZ}$ beads for U(VI) ions. (U(VI) concentration: 200 mg/L, Volume: 20 mL, pH: 6.0, Temperature: 25 °C, Stirring rate: 200 rpm)



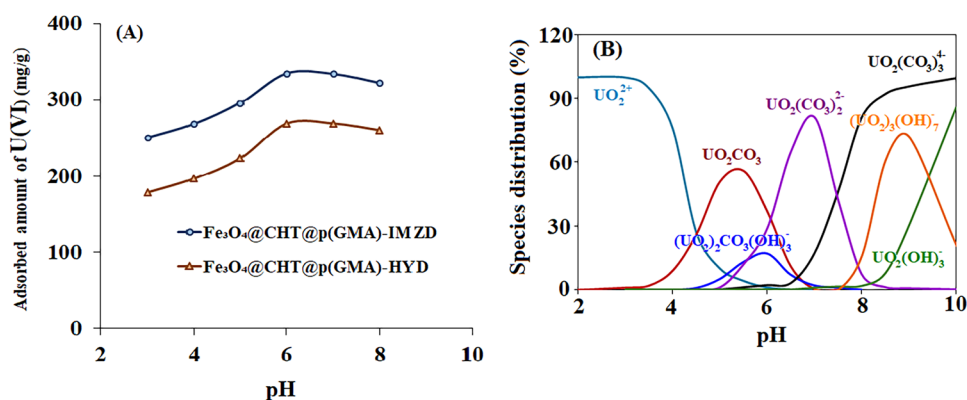
the hydrazone and imidazole groups of the adsorbents are 7.0 and 10.2, respectively. Thus, at the studied pH range, both adsorbent's surface is positively charged [45]. Thus, the surface of both adsorbents is protonated and positively charged at the studied pH range. At $\text{pH} \leq 5.0$, uranyl ions species are positively charged predominantly in the form of UO_2^{2+} ; thus, the electrostatic repulsion of both adsorbate and binding sites could prevent U(VI) ions from approaching the active binding center of the adsorbent, resulting in a decrease in the adsorption capacity of both adsorbent toward U(VI) ions. Whereas at $\text{pH} \geq 5.5$, negatively charged uranyl ionic species such as $\text{UO}_2(\text{CO}_3)_2^{2-}$, $\text{UO}_2(\text{CO}_3)_3^{4-}$, $\text{UO}_2(\text{OH})^{3-}$, $(\text{UO}_2)_3(\text{OH})_{10}^{4-}$ and $(\text{UO}_2)_3(\text{OH})_7^{7-}$ can be formed with increasing pH value of the medium. Therefore, the amount of adsorbed uranyl ions on the adsorbents can depend on the distribution of uranyl ion species at given pH values in the medium [46–49]. As a result, the adsorption capacity of $\text{Fe}_3\text{O}_4@\text{CHT}@p(\text{GMA})\text{-HYD}$ and $\text{Fe}_3\text{O}_4@\text{CHT}@p(\text{GMA})\text{-IMDZ}$ beads for uranyl ions were observed maximum at between pH 6.0 and 7.0. Due to the strong complexation of uranyl ions with the given above uranyl species, the main uranyl species were $\text{UO}_2(\text{CO}_3)_2^{2-}$ and $\text{UO}_2(\text{CO}_3)_3^{4-}$ at around pH 6.0 [1, 44]. Thus, the observed maximum adsorption capacities of uranyl ions on $\text{Fe}_3\text{O}_4@$

$\text{CHT}@p(\text{GMA})\text{-HYD}$ or $\text{Fe}_3\text{O}_4@\text{CHT}@p(\text{GMA})\text{-IMDZ}$ beads between pH 6.0 and 7.0 could be explained by the affirmative coordination conditions between uranyl ionic species hydrazone and imidazole groups. The uranium (VI) speciation diagram is also presented in Fig. 6B. As can be seen from the figure, the significant forms of uranyl ions between pH 6.0 and 7.0 are UO_2CO_3 , $(\text{UO}_2)\text{CO}_3(\text{OH})_3^-$ and $\text{UO}_2(\text{CO}_3)_2^{2-}$ [44]. From these observations, the binding interaction of uranium ions on $\text{Fe}_3\text{O}_4@\text{CHT}@p(\text{GMA})\text{-HYD}$ and $\text{Fe}_3\text{O}_4@\text{CHT}@p(\text{GMA})\text{-IMDZ}$ beads could be considered primarily as coordination complex formation, ion exchange interaction, and physical forces. It should be noted that $\text{Fe}_3\text{O}_4@\text{CHT}@p(\text{GMA})\text{-IMDZ}$ presented better adsorption capability for U(VI) ions than that of $\text{Fe}_3\text{O}_4@\text{CHT}@p(\text{GMA})\text{-HYD}$.

Effect of initial U(VI) ions concentration and isotherm models

Effect of initial concentration of U(VI) ions on adsorption performance of $\text{Fe}_3\text{O}_4@\text{CHT}@p(\text{GMA})\text{-DAMN}$, $\text{Fe}_3\text{O}_4@\text{CHT}@p(\text{GMA})\text{-HYD}$ and $\text{Fe}_3\text{O}_4@\text{CHT}@p(\text{GMA})\text{-IMDZ}$ beads was studied using the $\text{Fe}_3\text{O}_4@\text{CHT}@p(\text{GMA})\text{-DAMN}$ beads as a control system in the concentration range

Fig. 6 **A** Effect of pH on U(VI) adsorption on Fe_3O_4 @CHT@p(GMA)-HYD and Fe_3O_4 @CHT@p(GMA)-IMDZ beads. (Adsorbent dosage: 0.6 g/L, Temperature: 25 °C, Stirring rate: 200 rpm); **B** U(VI) ions speciation at 200 mg/L



of 1.0–400.0 mg/L. The maximum adsorption capacities of Fe_3O_4 @CHT@p(GMA)-DAMN, Fe_3O_4 @CHT@p(GMA)-HYD, and Fe_3O_4 @CHT@p(GMA)-IMDZ beads were determined to be 124.3, 328.4, and 434.7 mg U(VI)/g, respectively (Fig. 7). As seen in this figure, as the initial concentration of U(VI) ions in the medium increased, the adsorption capacities of the tested adsorbents increased. Because the initial concentration generates a driving force to overcome the counteraction of the mass transfer of uranyl ions between the adsorbent and medium [50–53], as the initial concentration of uranyl ions increases, the driving force increases, and thus, the adsorption capacity of the adsorbent will aggrandize. It should be noted that the modification Fe_3O_4 @CHT@p(GMA)-DAMN beads into hydrazine and imidazole groups increased the adsorption capacity by about 2.64 and 3.50 fold, respectively, compared to the parent compound.

Two adsorption isotherm models (i.e., Langmuir and Freundlich) were used to analyze the experimental data. The Langmuir model is described by the following:

$$C_e/q_e = (1/q_{\max})C_e + (1/q_{\max}b) \quad (2)$$

The Freundlich model is described by the following:

$$\ln q_e = (1/n)\ln C_e + \ln K_f \quad (3)$$

The correlation coefficients (R^2 values) were found to be 0.99 and 0.97 for Langmuir and Freundlich isotherm models, respectively, which means ($R^2 > 0.99$) that the adsorption of uranyl ions on the Fe_3O_4 @CHT@p(GMA)-HYD or Fe_3O_4 @CHT@p(GMA)-IMDZ beads well described by Langmuir isotherm model (Fig. 7B). The Langmuir isotherm model well described the possible interaction mechanisms of U(VI) ions with the functional groups of both adsorbent surfaces. Both tested

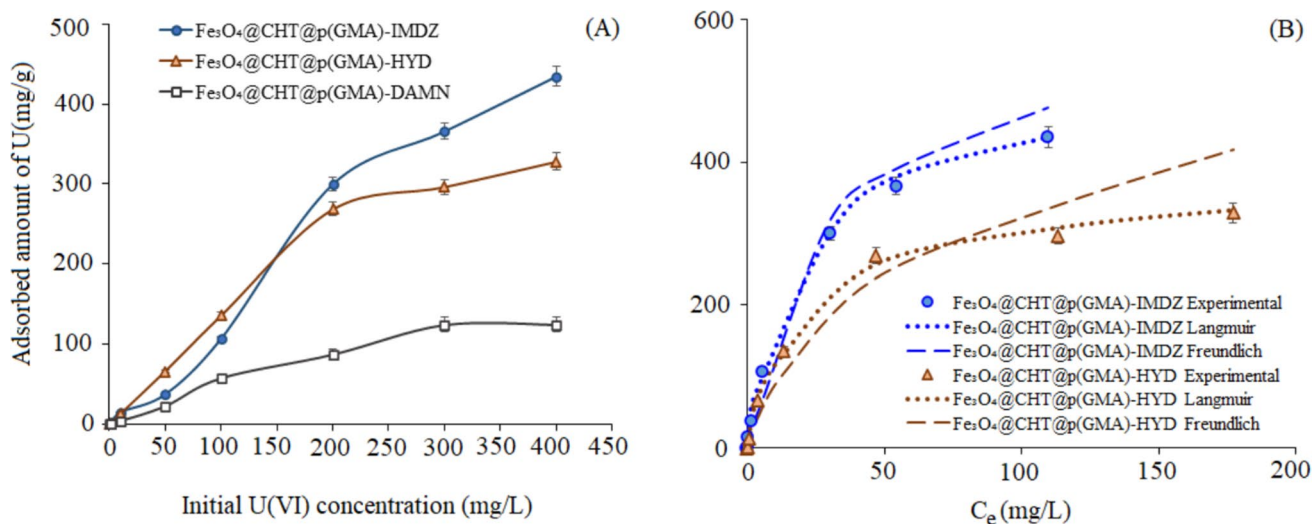


Fig. 7 The effect of initial concentration of U(VI) ions on the adsorption performance of Fe_3O_4 @CHT@p(GMA)-DAMN, Fe_3O_4 @CHT@p(GMA)-HYD or Fe_3O_4 @CHT@p(GMA)-IMDZ beads, (Solution volume: 20 mL, Temperature: 25 °C, Stirring

rate: 200 rpm) (**A**); Experimental, Langmuir and Freundlich isotherms plots for adsorption of uranium U(VI) on the Fe_3O_4 @CHT@p(GMA)-HYD and Fe_3O_4 @CHT@p(GMA)-IMDZ beads (**B**)

Table 1 Langmuir and Freundlich isotherm model parameters on the U(VI) removal using Fe₃O₄@CHT@p(GMA)-HYD and Fe₃O₄@CHT@p(GMA)-IMDZ beads

Adsorbents	Langmuir isotherm				Freundlich isotherm		
	q _{exp} (mg/g)	q _{max} (mg/g)	b × 10 ² (L/mg)	R ²	K _F (mg/g)	n	R ²
Fe ₃ O ₄ @CHT@p(GMA)-HYD	328.4	326.5	7.87	0.996	19.3	1.58	0.975
Fe ₃ O ₄ @CHT@p(GMA)-IMDZ	434.7	424.6	10.7	0.984	34.5	1.62	0.958

adsorbents had homogenous adsorption surfaces with hydrazone or imidazole functional adsorption sites. Thus, uranyl ions adsorption on the amine-rich ligands should follow through a dominant exchange mechanism. Thus, the adsorption of uranyl ions on both adsorbents was monolayer and the presence of homogeneous sites on their surface. The maximum theoretical U(VI) adsorption capacities of Fe₃O₄@CHT@p(GMA)-HYD or Fe₃O₄@CHT@p(GMA)-IMDZ beads were calculated to be 326.5, and 424.6 mg U(VI)/g, respectively. The theoretical adsorption capacities of U(VI) ions obtained from the Langmuir isotherm model were closely matched with the experimental adsorption capacities (Table 1).

Effect of contact time on adsorption of U(VI) ions on Fe₃O₄@CHT@p(GMA)-HYD and Fe₃O₄@CHT@p(GMA)-IMDZ beads

Figure 8 shows the effect of contact time on U(VI) ions adsorption onto Fe₃O₄@CHT@p(GMA)-HYD and Fe₃O₄@CHT@p(GMA)-IMDZ beads with an initial uranium ions concentration of 200 mg/L. The adsorption of U(VI) ions reached equilibrium at approximately 120 min and was slightly changed further during the continued contact period. Uranyl ions were adsorbed on the adsorbents by their functional groups until these groups were saturated;

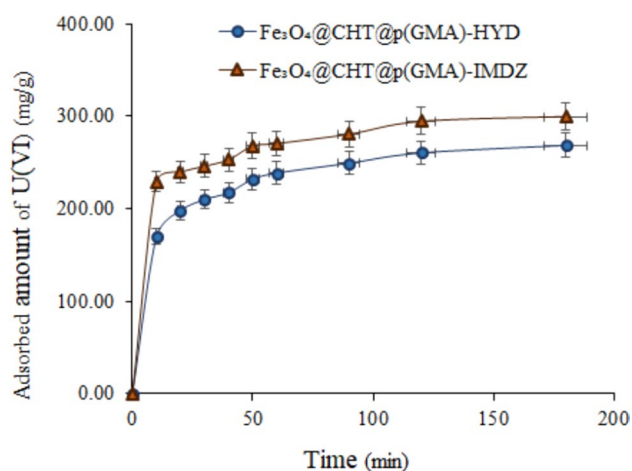


Fig. 8 Effect of contact time on the U(VI) removal using Fe₃O₄@CHT@p(GMA)-HYD and Fe₃O₄@CHT@p(GMA)-IMDZ beads. (Initial concentration: 200 mg/L, Adsorbent dosage: 0.6 g/L, pH: 6.0, Temperature: 25 °C, Stirring rate: 200 rpm)

then, the adsorption rate slowed down when approaching the adsorption equilibrium value. To evaluate the adsorption capacities of the Fe₃O₄@CHT@p(GMA)-HYD and Fe₃O₄@CHT@p(GMA)-IMDZ beads on uranyl ions at adsorption equilibrium.

The pseudo-first-order and pseudo-second-order kinetic models were used to analyze adsorption data. The equations for these models are:

$$\ln (q_e - q_t) = \ln q_e - k_1 t \quad (4)$$

$$t/q_t = 1/(k_2 q_e^2) + (1/q_e)t \quad (5)$$

According to the R² values, the adsorption of U(VI) ions onto Fe₃O₄@CHT@p(GMA)-HYD and Fe₃O₄@CHT@p(GMA)-IMDZ beads could be well described by the pseudo-second-order kinetic model. The findings revealed that the pseudo-second-order model exhibited the highest correlation coefficient for both adsorbents (Table 2). Additionally, the calculated adsorption capacities closely matched the experimental capacities. The second-order model assumes that the rate-limiting step may be chemisorption involving valence forces through the sharing or exchange of electrons between the functional groups of the adsorbent and the adsorbate.

Removal of U(VI) from synthetic and real seawater

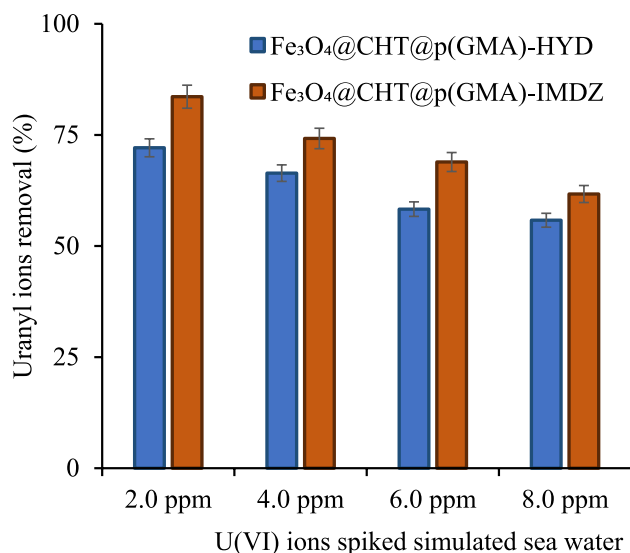
Fe₃O₄@CHT@p(GMA)-HYD and Fe₃O₄@CHT@p(GMA)-IMDZ beads were used to enrich uranyl ions from synthetic seawater. The former consisted of the following ingredients in g/1.0 L distilled water: NaCl: 24.5, MgCl₂ 5.20, CaCl₂: 1.2, KCl: 0.70, and CuCl, NiCl, and PbCl each 20 mg/L, with varying concentrations of uranium (i.e., 2.0, 4.0, 6.0, and 8.0 mg/L) and spiked with various amounts of uranyl ions. The performance of the tested two adsorbents in the synthetic seawater samples is shown in Fig. 9. The removal efficiencies of U(VI) ions by Fe₃O₄@CHT@p(GMA)-HYD were 72.1, 66.4, 58.3 and 55.8%, and for Fe₃O₄@CHT@p(GMA)-IMDZ were 83.6, 74.2, 68.9, and 61.7%, respectively, in U(VI) ions uranium spiked synthetic seawater with U(VI) concentration of 2.0, 4.0, 6.0, and 8.0 mg/L, after 3.0 h contact period. The enrichment of uranyl ions from synthetic seawater using Fe₃O₄@CHT@p(GMA)-HYD and Fe₃O₄@CHT@p(GMA)-IMDZ beads demonstrated

Table 2 Kinetic model parameters on the U(VI) removal using $\text{Fe}_3\text{O}_4@$ CHT@p(GMA)-HYD and $\text{Fe}_3\text{O}_4@$ CHT@p(GMA)-IMDZ beads

Adsorbent	Pseudo-first order model				Pseudo-second order model			
	$q_{e,\text{exp}}$ (mg/g)	q_e (mg/g)	k_1 (min ⁻¹)	R^2	q_e (mg/g)	k_2 (g mg ⁻¹ min ⁻¹)	R^2	
$\text{Fe}_3\text{O}_4@$ CHT@p(GMA)-HYD	328.4	249.2	4.72	0.921	332.5	3.42	0.999	
$\text{Fe}_3\text{O}_4@$ CHT@p(GMA)-IMDZ	434.7	235.0	4.57	0.966	427.6	4.11	0.999	

high removal performance in the samples spiked with varying amounts of uranyl ions. This result indicates that the synthesized adsorbents could be used for the recovery of uranyl ions from complex environment samples such as seawater, which contains various ionic species. The removal of uranyl ions by $\text{Fe}_3\text{O}_4@$ CHT@p(GMA)-HYD and $\text{Fe}_3\text{O}_4@$ CHT@p(GMA)-IMDZ beads decreased with increasing spiked amounts of U(VI) ions in the medium, and the percent removal efficiency of U(VI) ions decreased for adsorbents from 72.1 to 55.8% and from 83.6 to 61.7%, respectively.

The removal of the uranyl ions from the Black Sea and Mediterranean Sea Waters using $\text{Fe}_3\text{O}_4@$ CHT@p(GMA)-HYD and $\text{Fe}_3\text{O}_4@$ CHT@p(GMA)-IMDZ were also studied in a batch system, and the removal performance of $\text{Fe}_3\text{O}_4@$ CHT@p(GMA)-HYD was found to be 62.2 and 18.4%, and $\text{Fe}_3\text{O}_4@$ CHT@p(GMA)-IMDZ was 87.9 and 24.1%, respectively. It should be remembered that the Black Sea Water has a higher amount of uranium concentration than the Mediterranean Sea Water.

**Fig. 9** The effect of uranyl ions concentration on the removal performance using **a** $\text{Fe}_3\text{O}_4@$ CHT@p(GMA)-HYD and **b** $\text{Fe}_3\text{O}_4@$ CHT@p(GMA)-IMDZ beads from synthetic seawater samples

Adsorption mechanisms of U(VI) ions on $\text{Fe}_3\text{O}_4@$ CHT@p(GMA)-HYD and $\text{Fe}_3\text{O}_4@$ CHT@p(GMA)-IMDZ beads

Electrostatic and ion-exchange interactions can elucidate the adsorption mechanism of U(VI) ions by the $\text{Fe}_3\text{O}_4@$ CHT@p(GMA)-HYD and $\text{Fe}_3\text{O}_4@$ CHT@p(GMA)-IMDZ beads. The hydrazone groups are the primary functional binding sites of the $\text{Fe}_3\text{O}_4@$ CHT@p(GMA)-HYD adsorbent. On the other hand, the amount of nitrogen content of $\text{Fe}_3\text{O}_4@$ CHT@p(GMA)-IMDZ beads was further increased by incorporating ethylenediamine on the hydrazone structure to get obtain imidazole groups. These amine-rich groups have a high affinity toward metal ions. Both adsorbents adsorbed U(VI) ions with high affinity during adsorption, as observed from the above experimental results. As evidence, EDXS data were obtained from U(VI) ions laden $\text{Fe}_3\text{O}_4@$ CHT@p(GMA)-HYD and $\text{Fe}_3\text{O}_4@$ CHT@p(GMA)-IMDZ beads. The SEM micrographs, energy-dispersive-X-ray spectrum (EDXS), and SEM-EDX mapping analysis data for $\text{Fe}_3\text{O}_4@$ CHT@p(GMA)-HYD and $\text{Fe}_3\text{O}_4@$ CHT@p(GMA)-IMDZ beads are presented in Fig. 10 (partly A, B, C and D, E, F, respectively). As can be seen from Fig. 10, the energy dispersive X-ray spectrum and elemental mappings were obtained to evaluate the elemental distribution of adsorbents. As observed from this figure, the uniform distribution of hydrazone and imidazole major functional binding groups on the respective adsorbent resulted in uniform adsorption of uranyl ions on both adsorbent surfaces. Thus, it can be supposed that the data attained were clear evidence of the adsorption of uranyl ions on both adsorbent beads. Furthermore, as given in Fig. 2, the FT-IR spectrum of $\text{Fe}_3\text{O}_4@$ CHT@p(GMA)-HYD beads displayed peaks at around 3344 cm⁻¹ and 1640 cm⁻¹, corresponding to -NH₂ stretching vibration and N-H deformation, respectively, indicating the presence of amine groups. The FT-IR spectrum of the $\text{Fe}_3\text{O}_4@$ CHT@p(GMA)-IMDZ had a peak at 3311 cm⁻¹ for N-H stretching vibration, and the peaks at 1528 cm⁻¹ and 1385 cm⁻¹ were observed for -NH₂ and -CN stretching vibrations, confirming the formation of imidazole groups onto the HYD-modified beads after reaction with ethylene diamine. Therefore, these functional groups should be responsible for the electrostatic and ion-exchange

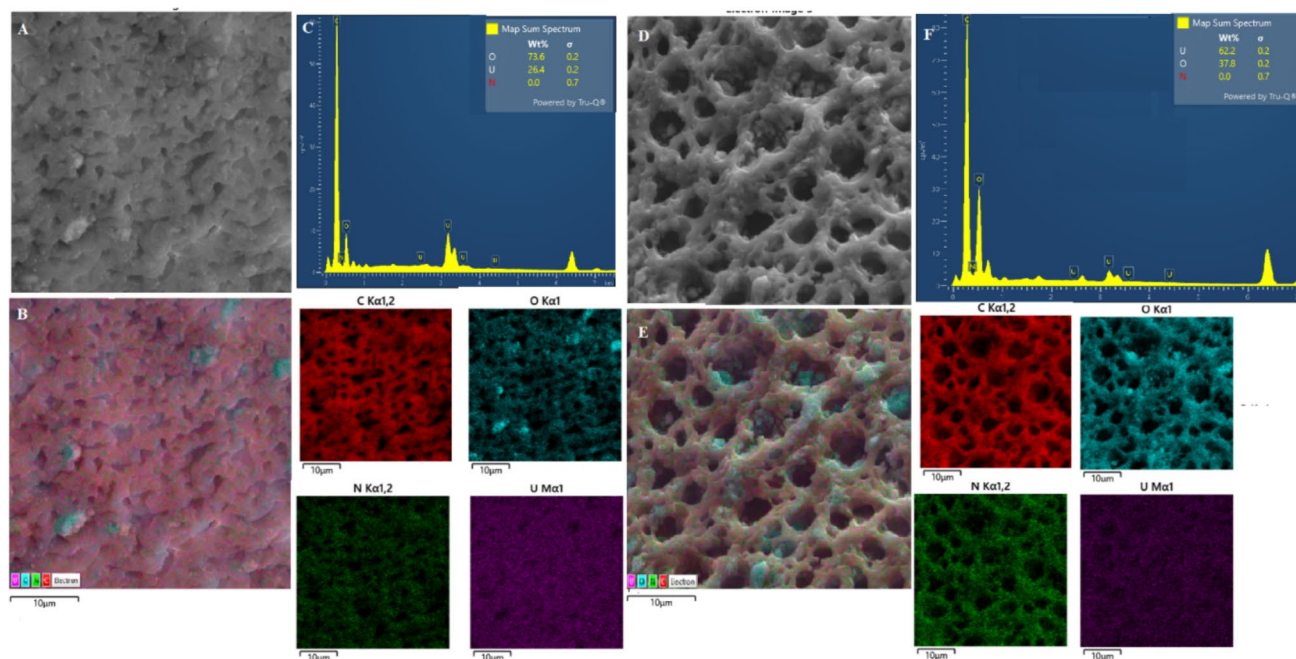


Fig. 10 SEM image, SEM–EDX mapping analysis and Energy-Dispersive-X-ray Spectrum of $\text{Fe}_3\text{O}_4@\text{CHT}@p(\text{GMA})\text{-HYD}$ (partly, A, B, C) and $\text{Fe}_3\text{O}_4@\text{CHT}@p(\text{GMA})\text{-IMDZ}$ beads (partly, D, E, and F), respectively

interactions and affect the adsorption performance of both adsorbents for U(VI) ions.

Competitive metal ions adsorption studies

To assess the adsorption selectivity of $\text{Fe}_3\text{O}_4@\text{CHT}@p(\text{GMA})\text{-HYD}$ or $\text{Fe}_3\text{O}_4@\text{CHT}@p(\text{GMA})\text{-IMDZ}$ beads adsorption experiments were performed in the binary competitive systems with U(VI)/Cu(II), U(VI)/Ni(II), and U(VI)/Pb(II) ions. In the binary and single systems, the adsorption studies of uranyl ions with these metal ions were achieved at pH 6.0 and with a concentration of metal ions 20 mmol/L. Figure 11 shows the amounts of adsorbed metal ions on the $\text{Fe}_3\text{O}_4@\text{CHT}@p(\text{GMA})\text{-HYD}$ and $\text{Fe}_3\text{O}_4@\text{CHT}@p(\text{GMA})\text{-IMDZ}$ beads in the single and binary metal ions systems. As observed from this figure, the presence of Cu(II), Ni(II), and Pb(II) reduced the adsorption capacities of the adsorbents for uranyl ions to some extent. The reduction in U(VI) adsorption capacity of $\text{Fe}_3\text{O}_4@\text{CHT}@p(\text{GMA})\text{-HYD}$ in the presence of Cu(II), Ni(II), and Pb(II) was 8.6, 6.3, and 10.0%, respectively. At the same time, the adsorption capacity of $\text{Fe}_3\text{O}_4@\text{CHT}@p(\text{GMA})\text{-IMDZ}$ for U(VI) was reduced by about 11.7, 28.9 and 25.6% in the presence of co-existence metals (i.e., Cu(II), Ni(II) and Pb(II)), respectively. The reduction in U(VI) adsorption capacity of $\text{Fe}_3\text{O}_4@\text{CHT}@p(\text{GMA})\text{-HYD}$ in the presence of Cu(II), Ni(II), and Pb(II) was 8.6, 6.3, and 10.0%, respectively. The adsorption capacity of

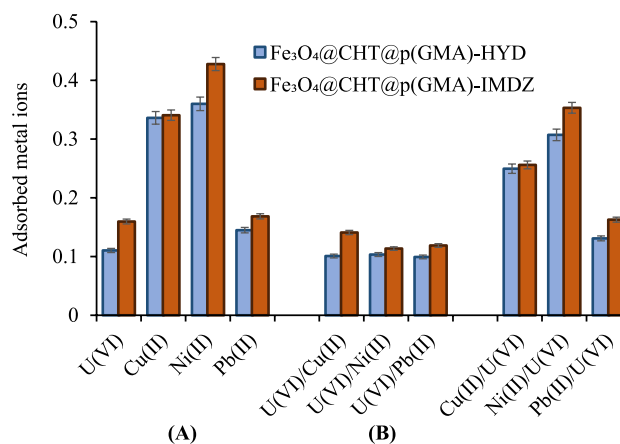


Fig. 11 Uranium competitive adsorption using $\text{Fe}_3\text{O}_4@\text{CHT}@p(\text{GMA})\text{-HYD}$ and $\text{Fe}_3\text{O}_4@\text{CHT}@p(\text{GMA})\text{-IMDZ}$ beads: **A** The adsorbed amounts of various metals from a single metal ion solution, **B** The adsorbed amount of U(VI) ions from in the presence of coexisting metal ions, and **C** The adsorbed amount of coexisting metal ions in the presence of U(VI) ions

$\text{Fe}_3\text{O}_4@\text{CHT}@p(\text{GMA})\text{-IMDZ}$ for U(VI) was reduced by 11.7, 28.9 and 25.6% in the presence of co-existence metals (i.e., Cu(II), Ni(II) and Pb(II)), respectively. The adsorption capacity of the $\text{Fe}_3\text{O}_4@\text{CHT}@p(\text{GMA})\text{-HYD}$ was less effective toward U(VI) ions in the presence of competitive metal ions compared to $\text{Fe}_3\text{O}_4@\text{CHT}@p(\text{GMA})\text{-IMDZ}$ beads.

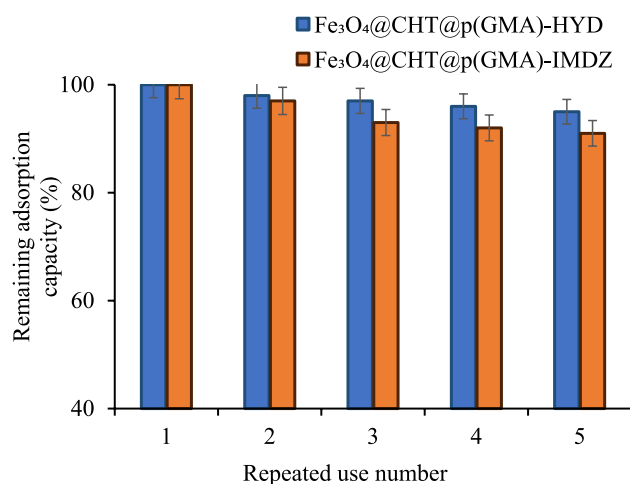


Fig. 12 Adsorption/desorption studies of Fe₃O₄@CHT@p(GMA)-HYD and Fe₃O₄@CHT@p(GMA)-IMDZ beads

Removal of U(VI) ions from seawater in a continuous system

Many continuous systems have been reported, such as stirred tank reactors, fluidized beds, and packed bed reactors. These continuous systems may exclude additional operational processes, such as filtration and centrifugation that require extra operational costs. Among these systems, the operation of a packed bed reactor may be easiest compared to another reactor system. The removal percentage of U(VI) in continuous systems using the Fe₃O₄@CHT@p(GMA)-IMDZ beads was investigated. Determination of the adsorption of U(VI) in a continuous system can be important for understanding the dynamic behavior of the adsorption process. Therefore, the Fe₃O₄@CHT@p(GMA)-IMDZ beads were operated in a packed bed reactor and used to remove U(VI) ions from the Black Sea and Mediterranean Sea waters. The initial concentrations of U(VI) in the reactor feed solutions were 9.4 and 5.5 pbp for the Black Sea and Mediterranean Sea waters, respectively. After a 24 h continuous operation of the reactor with a sample of sea waters, as described above, the

percentages of U(VI) ions removed from the Black Sea and Mediterranean Sea waters were determined to be 79.2% and 65.4%, respectively. No changes in the mechanical strength of the magnetic spheres were observed during the reactor's operation.

Regeneration and repeated use

For removing pollutants from wastewater, the ability to regenerate the adsorbent is a crucial economic consideration. Therefore, U(VI) ions laden Fe₃O₄@CHT@p(GMA)-HYD or Fe₃O₄@CHT@p(GMA)-IMDZ beads were desorbed with a dilute nitric acid solution (HNO₃, 50 mM). The desorption series was repeated five times using the same magnetic beads. As can be seen in Fig. 12, Fe₃O₄@CHT@p(GMA)-HYD or Fe₃O₄@CHT@p(GMA)-IMDZ beads continued a stable adsorption ratio of approximately 95 and 91%, respectively, and the adsorption efficiencies of adsorbents for uranyl ions were merely decreased by about 5.0 and 9.0%.

Comparison of the prepared Fe₃O₄@CHT@p(GMA)-HYD and Fe₃O₄@CHT@p(GMA)-IMDZ beads with earlier studies

It is essential to compare the adsorption efficiency of new adsorbents with present adsorbents for environmental applications. From Table 3, the adsorption capacities of the Fe₃O₄@CHT@p(GMA)-HYD and Fe₃O₄@CHT@p(GMA)-IMDZ beads for the uranyl ions can be compared to the reported works in the literature [14, 31, 32, 37, 38, 54–57]. The observed high adsorption efficiencies of the as-prepared adsorbents can result from electrostatic and coordination complex interactions between the hydrazone and imidazole groups of each adsorbent and U(VI) ions. As presented above, the imidazole groups were more reasonable for removing U(VI) ions from the medium. The high adsorption performance of the imidazole groups for uranyl ions can be a result of the specific interaction of this functional group with uranyl ions.

Table 3 Comparison of the adsorption capacities of adsorbents for uranyl ions

The adsorbent	Q _{max} (mg/g)	References
MCM-41- amidoxime or carboxyl groups	296.7/442.3	[14]
Layered double Zn/Al hydroxide-hexacyanoferrate(II) ions	156.7	[31]
Fe ₃ O ₄ coated by Zn and Al layered double hydroxide	268.7	[32]
Aromatic-based porous polymer-phosphonic acid	222.7	[37]
MGO@Si modified with amidoxime groups	641.4	[38]
Hydrazide grafted-magnetite chitosan derivative	368.9	[54]
Zr-MOF on functionalized graphene oxide	436.0	[55]
Polyaniline grafted magnetic carboxymethylcellulose beads	386.5	[56]
Amidoxime-Fe ₃ O ₄ @ TiO ₂ particles	277.3	[57]
Fe ₃ O ₄ @CHT@p(GMA)-HYD/Fe ₃ O ₄ @CHT@p(GMA)-IMDZ	328.4 / 434.7	In this work

Conclusion

In this study, magmatic chitosan beads were prepared via the phase inversion method and functionalized with diaminomaleonitrile ligand. The nitrile groups of the ligand were reacted with hydroxylamine and hydrazine hydrate, and the hydrozane and imidazole groups were obtained on the surface of the adsorbent and called “Fe₃O₄@CHT@p(GMA)-HYD or Fe₃O₄@CHT@p(GMA)-IMDZ beads, respectively. The prepared adsorbents were characterized using FT-IR, SEM, VSM, and BET methods. The prepared Fe₃O₄@CHT@p(GMA)-HYD or Fe₃O₄@CHT@p(GMA)-IMDZ beads were used for uranyl ions recovery from the aqueous medium, synthetic seawater in a batch system, and real seawater from the continuous system. The maximum adsorption capacities of the Fe₃O₄@CHT@p(GMA)-HYD or Fe₃O₄@CHT@p(GMA)-IMDZ beads were determined as 328.4 and 434.7 mg/g at pH 6.0, respectively. After five adsorption/desorption cycles, the uranium adsorption capacity was more than 87% of the initial adsorption capacity. The theoretical U(VI) ions adsorption capacity obtained from the Langmuir isotherm model was very near the experimental U(VI) ions adsorption capacity of the tested adsorbents. According to the tested kinetic models, the adsorption kinetics data of U(VI) ions on both adsorbents fitted the pseudo-second-order kinetics, which proposed that the chemisorption was the rate-limiting step. This research provides valuable insights into the potential of Fe₃O₄@CHT@p(GMA)-HYD or Fe₃O₄@CHT@p(GMA)-IMDZ beads for effective removal of trace amount uranyl ions from synthetic seawater and real seawater.

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Declarations

Conflict of interest The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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